

BUILDING BUSINESS READINESS FOR QUANTUM COMPUTING

KEY BARRIERS AND SUPPORT MECHANISMS

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Building business readiness for quantum computing

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Abstract

This paper examines how businesses are preparing to use quantum computing for commercial applications. It defines quantum readiness as an exploratory, capability-building process and describes the practical steps firms take to develop it. The study draws on the experience of public and private organisations based in 10 countries and supporting business engagement with quantum computing. The analysis identifies four key barriers: limited technological maturity; unclear use cases and business implications; high costs of access and staff training; and shortages of talent combining quantum computing expertise with industry-specific knowledge. The paper also examines how these barriers are being addressed through complementary support mechanisms, including networking and collaboration platforms, business advisory services, technology extension services, grants for business research and development (R&D), and stakeholder consultations. The findings provide actionable insights for policymakers and industry actors on how to strengthen readiness efforts and enable wider business engagement with quantum computing.

Keywords: quantum computing, business readiness, technology adoption, technology ecosystem, public-private support mechanisms, business advisory services, technology extension services, digital transformation.

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Executive summary

Quantum computing is an emerging paradigm in information processing that leverages the principles of quantum mechanics to solve certain complex problems that are difficult or lie beyond the reach of classical computers. While this technology is still at an early stage of development, potential commercial applications span a range of business sectors, including pharmaceuticals, advanced materials, energy, finance and transportation. As pioneering firms begin to explore the potential business value of quantum computing, the question emerges of how businesses can prepare for a technology that is both highly promising and highly uncertain.

This paper examines how firms can build quantum readiness, i.e. the strategic and operational preparations needed to evaluate, adopt and integrate quantum computing as the technology advances towards maturity. In this paper, quantum computing is treated as a long-term commercialisation effort: the value of quantum readiness does not lie in near-term production deployments but rather in early exposure and internal capacity to adapt as the technology matures. Drawing on the literature, on recent survey evidence and on interviews with 16 public and private organisations across 10 countries, the analysis identifies barriers that hinder readiness and the support mechanisms addressing these barriers.

Firms typically build readiness incrementally, starting with low-cost awareness and exploration. Early activities include participating in events and training sessions with technology providers and experts to learn about emerging opportunities, as well as organising workshops to explore potential applications. More advanced readiness requires investing in skills, infrastructure and partnerships to identify industry-specific use cases, assess potential business value and explore how quantum computing could be embedded in day-to-day operations.

In practice, companies face significant barriers: (i) insufficient technological maturity, which complicates pilots and feasibility studies; (ii) limited awareness and weak understanding of business implications; (iii) the costs of accessing quantum computing resources, funding research and development (R&D), and training staff, which are particularly burdensome for small- and medium-sized enterprises (SMEs); and (iv) shortages of talent combining quantum computing expertise and domain knowledge. As a result, readiness efforts tend to remain concentrated in a narrow group of large, R&D-intensive firms, while many others recognise the importance of quantum computing but delay action in practice. This risks a divide between early adopters and the wider digital economy, with uneven progress across firm sizes, sectors and countries.

Different kinds of public and private organisations help firms navigate readiness obstacles, including public research organisations, research and technology infrastructures, industry associations, technology providers and consulting firms, and research and innovation agencies. They do so through five main complementary support mechanisms:

- **Networking and collaboration:** platforms that connect researchers, technology providers, end users and other stakeholders to share information, track technological progress and identify market opportunities and potential partners.
- **Business advisory services:** services that familiarise firms with quantum computing, highlight industry-relevant cases and support firms in assessing their internal capabilities and gaps, while also providing training and strategic guidance on workforce and infrastructure needs.

- **Technology extension services:** services that bridge the gap between research and commercialisation by providing access to specialised facilities, equipment and expertise, and support for collaborative R&D for engaging with quantum computing and running proofs-of-concept.
- **Grants for business R&D:** funding that lowers the cost of early experimentation with quantum computing applications, while encouraging co-investment and collaboration with research and industry partners.
- **Stakeholder consultation:** ongoing engagement and exchanges with industry actors to align support measures with business needs, making readiness efforts more targeted and effective.

Looking ahead, building readiness will require expanding skills pipelines, strengthening industry-academic partnerships and ensuring that support mechanisms reach a wider range of firms. Hybrid approaches that combine quantum computing, artificial intelligence and high-performance computing are likely to serve as practical entry points for early commercial applications.

More broadly, policy support should evolve in tandem with the technology, balancing near-term exploratory pilots with forward-looking support in software, interoperability and workforce development. These measures can widen quantum readiness and bolster competitiveness and innovation across digital economies.

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Building business readiness for quantum computing: Key barriers and support mechanisms

Introduction

Quantum information technologies exploit the unique properties of atoms, photons and other particles to offer capabilities in gathering, processing and transmitting information beyond the reach of classical information and communication technologies (OECD, 2025^[1]). In particular, quantum computing is emerging as a new paradigm that is expected to solve complex problems that are challenging or impossible for today's classical computers. While the technology is not yet mature for wide commercialisation, businesses are beginning to explore its potential as a source of long-term competitive advantage.

This paper explores the key question of how companies can prepare for a technology that is both highly promising and highly uncertain. Section 1 introduces quantum computing, its potential applications and the concept of quantum readiness along with its main dimensions. Section 2 examines the roles of public and private organisations in helping develop quantum readiness. Section 3 identifies key barriers that companies face and approaches to overcoming them. Section 4 concludes with reflections on how readiness efforts may evolve and what this implies for industry actors and policymakers.

1. Towards quantum advantage and commercial applications

Quantum computing is an emerging approach to information processing that harnesses the principles of quantum mechanics to perform computations in fundamentally new ways. Unlike classical computers, which use binary digits, or “bits,” to represent either 0 or 1, quantum computers use qubits, which can represent multiple states simultaneously due to a quantum phenomenon known as superposition. Moreover, entanglement allows qubits to become intricately interconnected. These unique capabilities are expected to allow quantum computers to tackle certain complex problems that are difficult or even impossible for classical systems to solve.

In essence, qubits can hold and process far more information than classical bits: as more qubits are linked together, the number of possible configurations (the slots for encoding information) grows exponentially, allowing quantum computers to describe and process vast amounts of data. For example, a quantum computer with 300 qubits that performs as intended would have 2^{300} possible configurations, more than the number of particles in the known universe (Charley, 2022^[2]). Quantum operations then act on all of those possibilities simultaneously (a form of parallel data processing). For certain types of computational problems (see Box 1), this combination of richer information encoding and parallel processing can make quantum computers dramatically more powerful than classical machines.

Box 1. Quantum computing: Common misconceptions to avoid

Public discussions of quantum technologies sometimes exaggerate their current or near-term capabilities. The following clarifications can help set realistic expectations:

- **Exponential information does not automatically mean exponential speed:** Quantum computers can in principle represent very large amounts of information, but this does not translate into automatic or universal performance gains. Quantum technologies are expected to be useful in resolving only certain problems and progress remains gradual.
- **Superposition is not the same as doing millions of tasks at once:** Quantum states describe probabilities, not parallel calculations. Useful results still require carefully designed algorithms and repeated testing.
- **Quantum machine learning will not replace today's AI soon:** Noise, limited hardware and the difficulty of loading large volumes of data in quantum computers mean that, for most applications, artificial intelligence on classical computers will continue to be more effective for many years.
- **Claims of quantum advantage must be tested against classical approaches:** Several announcements of breakthroughs have been scaled back once faster or cheaper classical solutions were considered. Advantage is meaningful only when demonstrated under fair comparisons.
- **Timelines are uncertain:** Predictions about when quantum computing will be ready for broad commercial use vary widely. Policymakers and firms should treat such forecasts with caution and plan for multiple scenarios.

Source: OECD (2025^[1]).

Quantum advantage refers to the point in time when a quantum computer built and operated in practice outperforms a classical computer in solving a well-defined problem (Preskill, 2012^[3]). Quantum computing could provide a significant competitive advantage for businesses (Purohit et al., 2024^[4]) across various business activities and sectors (Box 2). However, the technology remains immature and has not demonstrated practical business value beyond the capabilities of classical computers (OECD, 2025^[5]). Despite significant theoretical promise, current quantum machines face major technical challenges, particularly the extreme sensitivity of qubits to environmental noise, which leads to high error rates and limits the duration and reliability of computations. Today's most advanced systems, such as superconducting chips, experience errors roughly every 100 to 1 000 two-qubit operations. Achieving industrial-scale applications will require reducing error rates by several orders of magnitude and scaling from hundreds to hundreds of thousands of qubits. Reaching the goal of fault-tolerant, large-scale quantum computers that can consistently outperform classical systems remains a future milestone. Nonetheless, progress in recent years in quantum error correction has strengthened the outlook for building reliable and scalable quantum computers. Researchers, large companies, start-ups and other actors worldwide are actively working to develop quantum computing and start realising this potential.

Box 2. Potential commercial applications of quantum computing across business activities

- **Medicine and healthcare:** Quantum computing could accelerate drug discovery and DNA sequencing, as well as contribute to more accurate patient outcome predictions.
- **Materials science:** Quantum computers could help model novel alloys and superconductors, guiding the development of stronger, lighter and more efficiently manufactured materials.
- **Energy:** Quantum simulation and optimisation may help deliver improved carbon-capture processes and energy grid operations.
- **Chemistry:** By simulating molecular reactions such as nitrogenase-based nitrogen fixation and photosynthesis, quantum computers could lead to new, more efficient and less costly processes for chemical manufacturing.
- **Finance:** Quantum algorithms could improve risk analysis, credit scoring and portfolio management, potentially optimising processes such as transaction settlements, arbitrage and crash forecasting.
- **Transportation and logistics:** Quantum computing could help design next-generation batteries and solve vast routing and scheduling problems, thereby increasing efficiency in vehicle fleets and cargo networks.

Source: OECD (2025^[1]).

Quantum information technologies operate fundamentally differently relative to classical information and communication technologies. As such, the private sector will need to invest considerable time and resources to prepare for the adoption of quantum technologies (Purohit et al., 2024^[4]). Anticipating the source of competitive advantage and potential disruption to their core businesses, several companies across industries are already exploring quantum computing to enhance operations, refine products and services, and strengthen their market position. While quantum computing remains in a pre-validation phase with multiple competing technology architectures, early exploration may nonetheless position firms to capture future benefits if viable approaches emerge.

Many companies across several sectors, particularly large firms, have started exploring use cases (Table 1), viewing quantum computing as a long-term strategic investment rather than an immediately applicable technology. A 2022 survey, covering 800 large organisations across 12 countries, found that about 23% were already working on or planning to work on quantum computing, sensing or communication applications (Capgemini, 2022^[6]). Among the firms pursuing quantum, close to 20% reported reaching implementation phase, i.e. integrating the technologies within their research and development (R&D) agenda and conducting experiments (13%), running limited-scale pilots or proofs-of-concept (3%) and achieving results from experimentation (3%). Around 70% of the organisations working on quantum underscored the need for early investment, citing long product development cycles and the time needed to integrate quantum technologies into their business processes. In 2024, the Boston Consulting Group identified more than 100 active quantum proof-of-concept projects among Fortune 500 companies, representing about USD 300 million in cumulative investment (Bobier et al., 2024^[7]). In this context, the 2024 revenue of quantum computing companies worldwide is estimated to range between USD 650-750 million (McKinsey, 2025^[8]).

Table 1. Selected examples of quantum computing use cases being explored by businesses

Use case	Firm	Industry	Application area	Algorithm class	Year
Aircraft loading optimisation	Airbus	Aerospace	Cargo transport	Optimisation	2022
Improving the design of composite materials	Boeing	Aerospace	Materials science	Optimisation	2023
Computational fluid dynamics in jet engines	Rolls-Royce	Aerospace	Manufacture	Simulation	2023
Molecular dynamics simulation	Toyota	Automotive	Materials design	Simulation	2023
Forecasting energy demand and renewable energy power output	EDF	Energy	Electricity distribution	Optimisation	2023
Portfolio optimisation	Goldman Sachs	Finance	Asset management	Optimisation	2023
Optimising arbitrage transactions	Hitachi	Finance	High-speed financial trading	Optimisation	2021
Molecular dynamics simulation	Boehringer Ingelheim	Pharmaceuticals	Drug development	Simulation	2021
Calculating the energy of molecules	Roche	Pharmaceuticals	Drug development	Simulation	2021
Communication service fault diagnosis	SoftBank	Telecommunications	Automation of network operations	Machine learning	2024

Source: (NEDO, 2025^[9]).

1.1 Building readiness for quantum computing

Quantum readiness refers to the strategic and operational preparations firms undertake to evaluate quantum computing and build the capabilities needed to adopt the technology once it reaches maturity (Yndurain, 2025^[10]). Rather than waiting, companies should proactively explore quantum computing and anticipate its business relevance. Firms that identify potentially meaningful impacts or opportunities should take steps to evaluate and align their capabilities and needs. This involves assessing existing and required skills, software and hardware infrastructure, and potential technology providers, as well as running small-scale pilots to build internal capabilities. Building quantum readiness represents a paradigm shift that requires companies to fundamentally rethink business problems and operations through a quantum lens. Ultimately, these efforts aim to enable companies to make informed, cost-effective decisions and gain a competitive edge. The following sections introduce the dimensions that are typically included in quantum computing readiness frameworks.

1.1.1 Strategic vision

Organisations should begin by building awareness and assessing the potential impact of quantum computing on their business strategy, evaluating the need to adopt a long-term, structured approach to adopting the technology. This involves assessing the sector exposure to quantum computing disruption or transformation, as well as defining acceptable business risk (Panteli, 2025^[11]). Companies may not perceive themselves as being sufficiently exposed to these risks, or they may lack the capacity to invest in building readiness. Quantum experimentation requires investment across several areas, as described in the subsections below. Firms must assess their need and capacity for sustained funding, involving multi-year R&D budgets dedicated to evaluating quantum hardware, algorithms and pilot projects (Boger, 2025^[12]). To justify such investments, enterprises need a compelling business case and a clear strategic objective. Without this strategic alignment, quantum projects risk remaining isolated experiments that fail to deliver quantum computing readiness. A well-defined strategic vision can help secure top management support and inform the need for a phased readiness roadmap, along with its component readiness activities (Teitsma, Ahmed and van Velzen, 2025^[13]).

1.1.2 Use case identification

Feasibility studies and use case analyses help organisations identify where quantum computing could deliver a competitive advantage. Firms should refrain from pursuing quantum computing solutions for problems that can be addressed efficiently by classical methods (Carrel-Billiard, Garrison and Dukatz, 2017^[14]). Instead, the strategic value lies in identifying and targeting computational bottlenecks or previously intractable challenges that, if solved, would unlock substantial business opportunities. This approach directs limited resources towards high-impact areas where quantum's distinct capabilities could deliver a demonstrable advantage. Beyond technical feasibility, organisations should assess quantum economic advantage, i.e. when a particular business problem can be solved more efficiently with a quantum computer than with a comparably priced classical machine (Stackpole, 2024^[15]). Organisations can conduct feasibility studies and use case analysis in stages to estimate potential business impacts, such as reduced operational costs and faster product development cycles (Panteli, 2025^[11]; Boger, 2025^[12]). Theoretical studies can evaluate how well quantum algorithms align with business problems. Pilot projects can test practical application using simulators or cloud-accessible quantum systems. More advanced assessments can then examine integration into existing systems and estimate costs, technical constraints and performance under real-world conditions.

1.1.3 Human capital

A shortage of skilled personnel is widely recognised as a barrier to quantum computing readiness (EY, 2022^[16]; IBM, 2023^[17]). Frameworks therefore often call for building a “quantum-ready workforce”, which involves both upskilling existing employees (e.g. through training programmes and workshops) and hiring new talent with relevant expertise (e.g. quantum scientists and quantum algorithm developers) (PwC, 2025^[18]). Early on, human capital investment may involve appointing an ambassador or forming a small quantum task force for quantum computing to disseminate knowledge and scout opportunities (Teitsma, Ahmed and van Velzen, 2025^[13]). As engagement deepens, dedicated quantum R&D groups or a “quantum computing office” could be established to coordinate internal efforts and external partnerships (Alsaman, 2023^[19]). Firms typically need a mix of different roles, including engineers with PhDs, solutions architects who can apply quantum algorithms to real-world problems and technicians who can maintain and operate systems, among others (OECD, 2025^[1]).

1.1.4 Infrastructure

Organisations taking steps to build readiness will need to access quantum computing resources and develop their technical infrastructure. Cloud-based services can meet most needs for small-scale experiments and prototypes, allowing companies to run pilot projects and gain hands-on experience (PwC, 2025^[18]). Such projects build internal know-how while quantum hardware continues to evolve, without incurring significant sunk costs. Organisations heavily invested in R&D may opt for on-premises quantum systems that enable hardware-level tuning for specific applications, safeguard data and use case confidentiality, and explore integration with their existing IT infrastructure (Liu, 2022^[20]). However, acquiring such hardware represents a considerable investment. The cost of a commercial quantum computer can range between USD 10-50 million, with additional software, operational and maintenance costs (Quantum Zeitgeist, 2024^[21]).

Infrastructure readiness also involves tracking the maturity of quantum hardware and algorithms. Organisations building readiness need to monitor technology roadmaps, hardware benchmarks (e.g. qubit counts and error rates) and algorithmic breakthroughs (IBM, 2023^[17]; Panteli, 2025^[11]). Moreover, quantum computing will not operate in isolation. It will need to be integrated with existing classical IT environments. A hybrid classical-quantum strategy is widely recognised as the most promising near-term approach, with the potential to provide initial business applications ahead of broader commercialisation of quantum

computing (OECD, 2025^[1]). Quantum resources will likely be accessed through hybrid models delivered primarily via cloud-native services (Panteli, 2025^[11]).

1.1.5 Ecosystem engagement

To prepare for quantum computing, organisations also need to understand the broader technology ecosystem (Yndurain, 2025^[10]). In the context of a rapidly evolving, highly specialised and costly field, most companies do not possess the internal capabilities to build readiness in isolation. Collaborating with external players (such as technology providers, research institutions, industry consortia and government actors) can help gain access to hardware, advanced algorithms, scarce specialised talent and market insights (Boger, 2025^[12]). Readiness frameworks often advise partnering with such actors, attending conferences, joining industry groups and co-developing proofs-of-concept and pilots (Teitsma, Ahmed and van Velzen, 2025^[13]). Government programmes also facilitate ecosystem connections. For instance, the UK's National Quantum Computing Centre works with businesses and academia to foster a quantum-ready economy, emphasising that companies are most likely to harness capabilities and gain first-mover advantage through such collaboration (EY, 2022^[16]). Japan's Ministry of Economy, Trade and Industry established the Global Research and Development Center for Business by Quantum-AI Technology at the National Institute of Advanced Industrial Science and Technology (AIST). The centre is developing several unique testbeds to support R&D for industrialising quantum technology. Key ecosystem engagement initiatives include creating use cases, developing the supply chain and building the business ecosystem, among others.

1.2 Building resilience for quantum computing

While quantum computing promises transformative benefits for businesses, it also poses serious digital security risks. A sufficiently powerful quantum computer could compromise cryptographic methods that are prevalently used today to ensure the security of practically all data transmitted online, including public or private sector data with strategic, financial or intellectual property value (OECD, 2025^[1]). In a survey of 32 leading science and industry experts in the field, approximately two-thirds estimated a 50% or greater likelihood of the arrival of a cryptographically relevant quantum computer within the next 15 years (Mosca and Piani, 2024^[22]). Quantum computing resilience refers to transitioning to new cryptographic solutions that are resistant to quantum attacks, like post-quantum cryptography and quantum key distribution, before such attacks materialise. Cybersecurity agencies and financial authorities are urging public and private organisations to begin implementing post-quantum cryptography.

Both readiness and resilience are essential activities for organisations to prepare for the quantum future, yet they require different kinds of business preparations (Box 3). Quantum resilience focuses on addressing the risks that future quantum computers pose to today's cryptographic systems. This involves identifying and inventorying critical data, assessing the use of vulnerable algorithms, conducting risk assessments and developing transition plans, among other key activities (Homeland Security, 2022^[23]) that are separate from commercial applications. For this reason, efforts around building resilience are kept outside the scope of this paper.

Several organisations interviewed as part of this work highlighted that quantum resilience, and particularly the transition to post-quantum cryptography, provides a concrete and urgent entry point for firms to engage with quantum computing. The risk of “store-now, decrypt-later” attacks is compelling organisations to act now, with compliance obligations and sectoral pressures (especially in finance) driving early investment. This urgency has created opportunities for supporting organisations to initiate dialogue with companies, often by framing roadmaps that begin with cybersecurity and then extend towards broader commercial applications. Some interviewees cautioned that readiness efforts may remain hidden within cybersecurity budgets and may not translate into visible pilots in computing. However, there was broad agreement that security concerns are establishing crucial channels for awareness-raising. In this sense, quantum

resilience can serve as a necessary first conversation that opens the door for companies to develop quantum readiness.

Box 3. Building resilience to quantum computing

Why organisations need to start now

Although the timeline for quantum computers capable of breaking current cryptographic methods remains uncertain and likely more than a decade away, cybersecurity agencies consider the threat to be already urgent. This is because vulnerable cryptographic systems are widely used in most digital communications. Additionally, encrypted data intercepted today could be decrypted in the future using quantum computing capabilities, a threat known as "store now, decrypt later" attacks. Moreover, transitioning to quantum-resilient cryptography is a complex and lengthy process that may take up to 20 years, given the scale of digital infrastructure and the number of actors involved.

Post-quantum cryptography

Post-quantum cryptography (PQC) refers to cryptographic methods designed specifically to withstand attacks from quantum computers. Unlike traditional encryption algorithms, which are vulnerable to quantum computing, PQC employs classical computing algorithms based on complex mathematical problems designed to be challenging even for quantum algorithms. PQC is intended for implementation on existing computing and networking infrastructures, facilitating widespread adoption and a smooth transition to quantum resilience. However, PQC methods require greater computational resources and higher energy budgets per key exchange than current cryptographic approaches. They may also present compatibility issues for older devices (e.g. Internet of Things).

Quantum key distribution

Quantum key distribution (QKD) is a quantum communication technology that aims to secure key exchanges by leveraging the laws of physics rather than relying on mathematical complexity. QKD could help ensure that any intercepted information is rendered unusable. In practice, however, the security of practical QKD systems is highly implementation dependent. QKD devices can introduce vulnerabilities that attackers may exploit, meaning that actual security depends on hardware and protocol implementations rather than physics alone. Moreover, relative to PQC, QKD requires costly, specialised hardware that complicates integration into existing infrastructure and imposes logistical challenges. QKD also provides only a partial cryptographic solution, as it does not supply source authentication and therefore still requires classical cryptographic mechanisms. The study and resolution of such vulnerabilities and practical challenges is an active area of research and development, and experts see QKD and PQC as complementary rather than mutually exclusive.

Sources: OECD (2024^[24]) and OECD (2025^[11]).

2. Supporting readiness for quantum computing

Although quantum computing is still in its early stages and poses significant financial risks, the promise of high returns has attracted private investors. However, the extended timelines to reach commercial viability may challenge the sustainability of such investment (Gibney, 2019^[25]). Governments are currently driving investment in these technologies, as they view them as a potential future productivity engine and source of industrial advantage. Public funding also helps governments have oversight on technology developments that involve risks to digital security or have defence implications (OECD, 2025^[5]). As of

September 2025, governments worldwide have announced an estimated USD 55.7 billion in quantum technology investments since 2013 (Qureca, 2025^[26]).

As discussed in the previous section, business efforts to build quantum readiness are primarily being led by large firms. Recent surveys indicate that most firms are unfamiliar with quantum computing's practical applications:

- In 2022, the consulting firm EY partnered with the United Kingdom's National Quantum Computing Centre to survey 501 executives across 11 sectors in the country on their companies' quantum readiness (EY, 2022^[27]). The results showed enthusiasm but limited preparedness: While 97% of these UK business leaders believe quantum computing will moderately or significantly disrupt their industry, only one-third of participating organisations had begun any strategic planning for readiness. More than 1 000 executives declined to complete the survey, citing insufficient familiarity with the technology.
- A 2023 survey of 200 data, analytics and innovation leaders in the financial services sector from 17 European and North American countries found that while 87% of respondents regard quantum computing as an opportunity, 73% had not identified any application where the technology could provide a real commercial advantage (Moody's, 2023^[28]). Reflecting this pattern, 87% of firms participating in the survey had no dedicated budget for quantum initiatives.
- In a 2025 survey of 2 600 global professionals in digital trust, cybersecurity, IT audit, governance and risk, 63% of respondents anticipate quantum computing to accelerate computational tasks or data analysis significantly, while 46% expect it will create revolutionary innovations (ISACA, 2025^[29]). Moreover, the majority of respondents (64%) expect the transformative potential of quantum computing to become widespread across their industries within the next decade. Despite this optimism, only 20% of respondents indicated that their organisations have formal plans in place to build quantum readiness.

Taken together, these studies reveal a paradox: Businesses acknowledge the importance of quantum computing in principle, yet their practical engagement is limited. Table 2 synthesises three recent surveys that gather data on barriers to building quantum readiness, drawing from diverse actors including quantum computing scientists, industry experts, financial sector leaders and quantum technology developers. Across these surveys, technology immaturity and unclear business value emerge as recurring concerns, as businesses recognise that quantum systems are yet to demonstrate an economic advantage. The QuEra (2025) survey highlights the practical hurdles that organisations confront once they begin exploring quantum computing, including high costs, scarce talent and, to a lesser extent, limited access to hardware. Canadian quantum technology developers (ISED, 2024^[30]), located upstream in the value chain, point to a lack of compelling, validated use cases and customer knowledge gaps.

Many public and private organisations have expertise in emerging technologies, such as research centres, technology providers, innovation hubs and industry associations. Such organisations play an essential role in ensuring that the benefits of new technologies can be realised more broadly across economies and sectors beyond a narrow group of early adopters (OECD, 2017^[31]). They help firms navigate the complexity, high costs and fast-paced evolution associated with advanced technologies, including by:

- Closing businesses' information gaps and providing access to infrastructure, supporting firms in identifying relevant applications, adapting technologies to their specific contexts, and integrating them into existing processes and business models; and
- Fostering collaborative experimentation among developers, users and intermediaries, accelerating learning and reducing uncertainty.

Over time, such support helps firms build the capabilities they need not only to adopt new technologies, but also to scale them and continuously improve their use. The following section explores the role of such organisations in supporting quantum readiness.

Table 2. Recent survey data on barriers to building quantum readiness

Source	(QuEra, 2025 ^[32])	(Moody's, 2023 ^[28])	(ISED, 2024 ^[30])
Survey year	2024	2023	2024
Focus	Quantum computing	Quantum computing	Quantum technologies
Sample	770 quantum computing experts, researchers and decision-makers from 50 countries, primarily in Europe (39.6% of responses), North America (35.6%) and Asia (14.9%).	200 data, analytics and innovation leaders in financial services and banking firms from 17 European and North American countries. - Respondents were affiliated with businesses generating annual revenues of USD 100 million or greater.	112 individuals in Canada's quantum sector developing quantum technologies, including 66 from industry and 46 academics. 46% of industry respondents conduct research and development in quantum computing.
Survey question	What challenges would or does your organisation face in adopting quantum computing?	What are the biggest operational challenges for the development of quantum computing in your organisation?	What is the main barrier, if any, preventing end users from adopting quantum solutions?
Question type	Multiple choice	Multiple choice	Single choice
Question sample	Users or potential users of quantum computing (12.7% of survey sample)	Full survey sample	Industry actors developing quantum technologies (50% of survey sample)
Main barriers reported	- High costs of implementation: 39.6% - Lack of skilled workforce: 40.6% - Unclear business value: 39.6% - Technology immaturity: 34.4% - Immaturity of algorithms: 28.1% - Limited access to hardware: 21.9% - Integration with existing systems: 17.7% - Competition with other priorities: 17.7% - Lack of internal buy-in: 15.6% - Security and data privacy: 15.6% - Regulatory compliance concerns: 9.4%	- Technology immaturity: 82% - Lack of C-level executive buy-in: 48% - General scepticism about the potential of quantum computing: 41% - Lack of investment: 37% - Integration with existing systems: 33% - Lack of skills in-house: 32% - Cybersecurity concerns: 22% - No access to partnerships with other expert institutions: 4%	- Use cases still being proven: 21.4% - Technology immaturity: 19.6% - Lack of knowledge: 19.6% - Technology costs: 8.9% - Risk aversion: 7.1% - No urgency/incentive for it: 5.4% - Other barriers: 14.3% - No barriers: 3.6%

2.1 Organisations helping firms build readiness for quantum computing

Between April and May 2025, the OECD Secretariat organised 16 structured interviews with public and private organisations across 10 countries that support companies in building quantum readiness. These organisations possess a wealth of technical and business knowledge derived from their extensive interaction with companies and other actors in quantum computing ecosystems. They were identified through the OECD Quantum Technologies Policy Database (OECD, 2025^[33]) and in discussions of the OECD Global Forum on Technology focus group of experts on quantum technologies. They have been selected to ensure regional variety and to capture different types of organisations. They can be classified under five types (Table 3): public research organisations, research and technology infrastructures, industry associations, technology providers and consulting companies, and research and innovation agencies.

The interviews were structured around three main sections. Participants were first asked about their organisation's role, the scale of their engagement in quantum readiness, and their strategies for identifying and reaching potential adopters of quantum computing. Interviews then explored perceptions of industry awareness and readiness, focusing on the barriers that hinder firms. The final section explored participants' views on the most effective forms of support currently available, as well as desired future developments in support mechanisms.

Table 3. Interviewed organisations helping firms build readiness for quantum computing

Type	Description	Role in knowledge diffusion	Organisations / programmes interviewed
Public research organisations	Organisations (e.g. universities, research centres) that conduct long-term research, often in specialised fields, to develop new knowledge and technologies, sometimes focusing on innovation and industry support.	Public research organisations generate much of the scientific and technological knowledge that later feeds business innovation. They also train researchers, promote collaboration and support the transfer of research into commercial applications.	- Quantum Computing Initiative (German Aerospace Centre), Germany - QuSoft, Netherlands
Research and technology infrastructures	Facilities, tools and resources, such as large-scale scientific equipment, specialised databases, supercomputing facilities and demonstration plants.	As they host specialised equipment, data and skills, research and technology infrastructures attract diverse stakeholders, fostering collaboration across disciplines and borders, and often anchor entire innovation ecosystems.	- National Physical Laboratory, United Kingdom - Qu-Test (Horizon Europe), European Union - Quantum computing initiative (German Aerospace Centre), Germany
Industry associations	Industry associations unite companies within specific sectors or around shared technologies to represent and advance their common interests.	Industry associations serve as a forum for exchanging ideas and information, including on emerging technologies.	- Quantum Strategic Industry Alliance for Revolution (Q-STAR), Japan - Quantum Economic Development Consortium (QED-C), United States - Quantum Technology and Application Consortium (QUTAC), Germany
Technology providers and consulting companies	Private entities that offer technical solutions, expertise and advisory services to support the development and adoption of technologies across industries.	They help deliver tailored software or hardware systems, offer strategic and operational guidance on innovation processes, or support digital and technological transformation through applied knowledge and tools.	- Microsoft, United States - Pasqal, France - Qureca, United Kingdom - ZirconTech, Uruguay
Research and innovation agencies	Public or publicly funded bodies that provide competitive grants and other forms of support to advance scientific knowledge, technological development and societal impact.	These agencies can create incentives for collaboration between science-industry stakeholders, promoting the commercialisation of research and the spread of technical knowledge throughout the economy.	- Commercialising Quantum Technologies Challenge (Innovate UK), United Kingdom - CSIRO, Australia - Critical Technologies Challenge Program (Department of Industry, Science and Resources), Australia - Grand Solutions initiative (Innovation Fund Denmark), Denmark - Quantum computing initiative (German Aerospace Centre), Germany - VDI Technologiezentrum, Germany

Sources: OECD elaboration drawing from interview data; (EC-OECD, 2025^[34]).

Interviews were conducted on record with consent to attribution. Their duration ranged between 30-60 minutes. A standard set of questions used to guide the interviews can be found in Annex A. Questions were tailored to each organisation based on information available online. This structured approach was designed to enable consistency across interviews as well as flexibility to explore specific organisational perspectives. Analysis of the data gathered in these interviews points to five main mechanisms through which the organisations support businesses building quantum readiness, described in Box 4: networking and collaborative platforms, business advisory services, technology extension services, grants for business R&D, and stakeholder consultation.

Box 4. Main support mechanisms for companies building quantum readiness

- **Networking and collaborative platforms** connect researchers, technology providers, end users and other stakeholders to share information on emerging use cases, technical knowledge and resources. They support companies in understanding quantum computing and monitoring technology developments, as well as in identifying market opportunities and potential partners.
- **Business advisory services** aid firms to familiarise themselves and engage with quantum computing. They include, for example, workshops that introduce key concepts and highlight relevant industry cases. Services can also help firms assess their capabilities and identify gaps in knowledge, infrastructure or workforce skills. They can also assist in upskilling employees through targeted training programmes and offer strategic guidance on future staffing needs.
- **Technology extension services** help bridge the gap between research and technology commercialisation. They can raise companies' capabilities in quantum computing by facilitating access to advanced and expensive hardware systems, equipment, facilities and specialised expertise, as well as by supporting collaborative R&D with research and industry partners. These services support firms in conducting feasibility studies and proofs-of-concept that explore business-specific use cases.
- **Grants for business R&D** ease the financial burden of investing in an emerging technology like quantum computing. The funding can help firms access quantum computing resources and support R&D investments needed to explore business-specific applications. Grants may require beneficiaries to co-invest, as well as collaboration with research or industry partners to foster knowledge exchange.
- **Stakeholder consultation**, while a less direct mechanism, plays an important role in shaping support measures. Tools such as expert panels and surveys can identify industry challenges, needs and opportunities, helping to ensure that these are reflected in policies. Consultations contribute to the design of more targeted and effective forms of support.

Source: OECD elaboration drawing from interview data and policy instrument descriptions from EC-OECD (2025^[34]).

Table 4 presents the support mechanisms provided by participating organisations. It illustrates how different types of organisations contribute to building readiness for quantum computing through complementary forms of support. The following section describes the main barriers identified by these organisations and how they are being addressed. Readers interested in a broader collection of policies supporting the development and uptake of quantum technologies may refer to OECD (2025^[33]).

Table 4. Mapping of support mechanisms by interviewed organisations

Jurisdiction	Organisation/programme	Networking and collaborative platforms	Business advisory services	Technology extension services	Grants for business R&D	Stakeholder consultation
Japan	Quantum Strategic Industry Alliance for Revolution (Q-STAR)					
United States	Quantum Economic Development Consortium (QED-C)					
Spain	Qureca					
Germany	Quantum Computing Initiative				see note	
Germany	Quantum Technology and Application Consortium (QUTAC)					
United Kingdom	Innovate UK					
United Kingdom	National Physical Laboratory					
France	Pasqal					
United States	Microsoft					
Uruguay	ZirconTech					
European Union	Qu-Test (Horizon Europe)					
Netherlands	QuSoft					
Australia	Critical Technologies Challenge Program (Department of Industry, Science and Resources)					
Denmark	Grand Solutions: Quantum Technologies (Innovation Fund Denmark)					
Germany	VDI Technologiezentrum					
Australia	Commonwealth Scientific and Industrial Research Organisation (CSIRO)					

Note: While Germany's Quantum Computing Initiative does not award grants, it funds companies through industry contracts (procurement of R&D services). Through these contracts, firms develop quantum hardware components or contribute domain-specific expertise relevant to applications. For example, Deutsche Bahn provided train optimisation use cases for testing on quantum hardware.

Source: OECD elaboration based on data gathered via interviews and desk research.

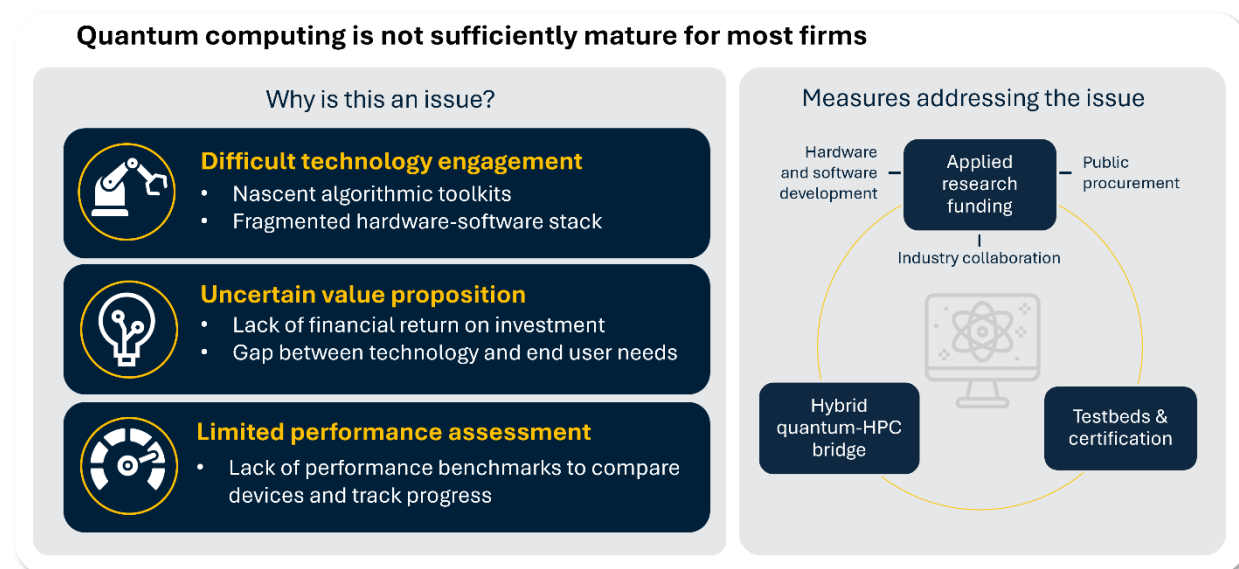
3. Key barriers to readiness and related support measures

3.1 Quantum computing is not a sufficiently mature technology

While interest and investment in this field are growing rapidly, quantum computing is not yet a sufficiently mature technology for widespread deployment. Existing hardware is limited to small-scale processors with high error rates, and there are multiple competing quantum computing hardware platforms (e.g. superconducting qubits, trapped ions, photonic systems) with no single one emerging as dominant. These limitations mean that most potential commercial applications of quantum computing, such as optimisation and machine learning, remain experimental and do not yet represent an economic advantage over classical methods. In addition, the emerging software ecosystem is fragmented, lacking standardised programming frameworks or mature algorithmic toolkits tailored to specific sectors. This context makes it

difficult for businesses to identify clear use cases or to justify large-scale investment in skills or infrastructure.

Infographic 1. Obstacle overview and related support measures



Experts working at the interface between technology developers and industrial users note a persistent disconnect between the two groups. As Claudius Klein, Senior Technology Consultant at [VDI Technologiezentrum](#), an innovation agency in Germany, explains, quantum computing SMEs and start-ups tend to focus on advancing the technology with public support, while large industrial end users expect ready-made practical solutions. Bridging this gap is essential, as quantum computing is not yet capable of delivering the kinds of industrial solutions that firms expect.

Interviewees noted a pattern where firms acknowledged the long-term significance of quantum computing but did not see immediate returns or pressing business applications that justified near-term investment. As a result, with often limited resources and R&D budgets, firms tend to prioritise more immediate technologies over quantum computing. As stated by an interviewee, quantum computing is not yet a top issue in boardrooms. Instead, companies are pooling resources towards more developed and commercially viable technologies, such as artificial intelligence. This is particularly evident in emerging economies,¹ where business sectors continue to lag in the adoption of digital technologies.

The main obstacles towards the maturity of quantum computing relate to engineering challenges rather than fundamental physics. However, the immaturity of the hardware and algorithmic landscape directly limits the return on investment that businesses can demonstrate by investing in quantum computing today. The complexity of the set of technologies (or “stack”) that underpins quantum computing is a related challenge. This stack includes hardware, control systems, software and interfaces needed to integrate quantum with existing high-performance computing and AI systems. At present, interoperability across different providers is challenging, leaving firms uncertain about how these components could work together in practice. Progress will require time, standardisation and international coordination to achieve compatibility and reduce complexity.

For many businesses, the lack of commercial applicability makes it challenging to justify investing in quantum readiness. Until quantum economic advantage becomes a near-term prospect, many enterprises will remain hesitant to invest in a technology that is not yet aligned with operational needs or business timelines. However, many companies with substantial R&D budgets invest early, recognising that building readiness takes several years and can create a lasting competitive advantage. Large corporations with the

resources to explore multiple emerging technologies often invest in quantum not because of immediate applications, but because of a concern about being left behind on the next transformative wave. This tendency reflects lessons learned from artificial intelligence, where many firms realised the cost of entering the field too late. In the case of quantum computing, some companies are determined to be early movers, even if the technology is still far from commercial deployment, to avoid being left behind should breakthroughs occur.

Businesses that are building readiness face significant frictions arising from the fragmentation of the quantum computing technology ecosystem. Multiple quantum hardware platforms are being developed in parallel, each with its own programming tools, languages, strengths and limitations. This diversity reflects a healthy and necessary phase of technological competition and experimentation. However, the coexistence of multiple, largely incompatible approaches across the full quantum computing stack, from hardware to middleware and software, makes it difficult for companies to navigate the ecosystem and plan long-term investments. For example, quantum software is not yet hardware-agnostic and remains tied to specific machines and architectures, making it challenging for potential end users to develop transferable skills and applications. The coexistence of multiple competing approaches generates uncertainty for potential adopters, who lack clear market signals about which hardware platforms are likely to prove most viable for their intended applications.

A further complication is the lack of standard benchmarks and performance metrics across different quantum systems. As hardware platforms are built on diverse modalities and optimised for different workloads, they are not straightforward to compare. Until benchmarking standards are better established, it will remain challenging for firms to evaluate the value and applicability of different systems. These dynamics discourage many firms from incorporating quantum readiness into their digital strategies, as it makes it difficult to scope requirements and plan for feasibility studies, staffing and infrastructure.

Governments are addressing these challenges through public investment in applied research activities that contribute to quantum computing's journey towards commercial maturity:

- **Applied research for hardware and software development:** Governments support applied research to accelerate the transition from basic quantum science to practical technologies, typically through collaborative grant programmes that fund partnerships between academia and industry. Denmark's [Grand Solutions programme](#) (Innovation Fund Denmark), for example, issues calls for collaborative projects within quantum technologies. The programme supports activities along the entire value chain, from early-stage foundational and applied research to projects demonstrating quantum technologies. While collaboration is a requirement, industrial participation is not mandatory. Given the immaturity of the field, the programme allows for academic-only partnerships. Projects may receive grants ranging from DKK 5-40 million (approximately EUR 650 000 to 5.5 million) and typically run for periods of 1-5 years. Eligible expenses include postdoctoral and PhD positions, equipment and materials for research teams developing quantum hardware and software prototypes. Germany's [Quantum Computing Initiative](#) (German Aerospace Centre) includes an applied research programme supporting the development of two main quantum computing hardware platforms: superconducting and trapped ions. Research projects also seek to create algorithms that can run on existing quantum computers and to develop commercial applications in fields such as materials science and logistics.
- **Public procurement of quantum computers:** Governments across many countries have financed the installation of quantum computers in public research centres to host early-generation machines, offering platforms for scientists and industry partners to test applications and build skills at a stage when private demand is still limited. Procurement helps sustain hardware vendors through long development cycles and demonstrates viability to private investors. Germany's Quantum Computing Initiative aligns procurement contracts with application projects where companies are paid to contribute data, domain knowledge or software expertise. This model aims

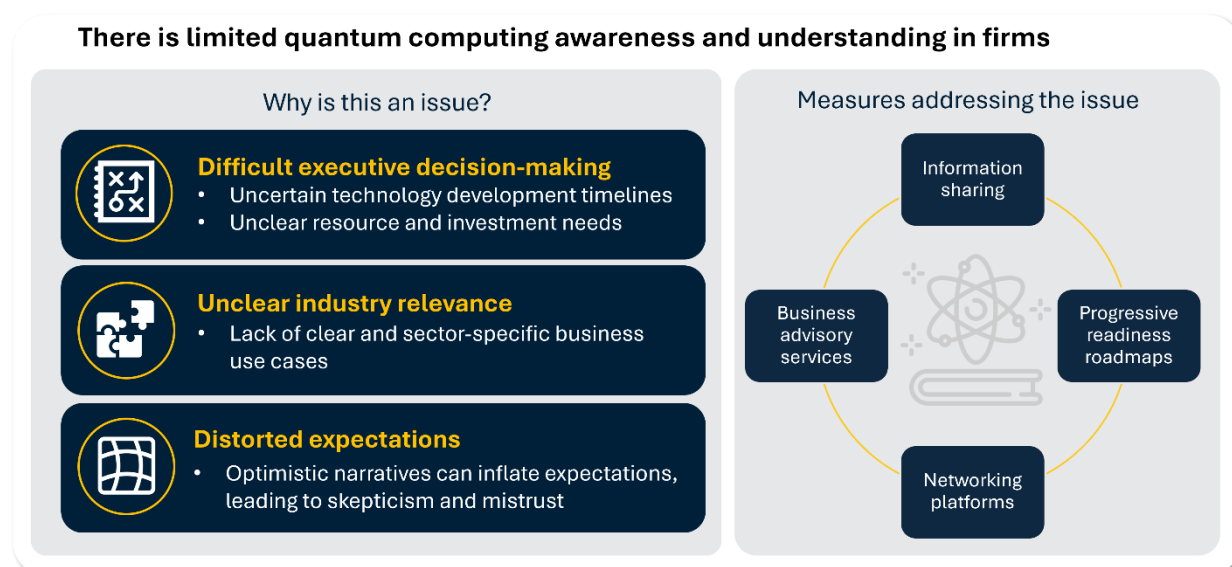
to ensure that procured systems are not idle research tools but are embedded in industry-relevant experiments, building bridges to commercial use cases.

- **Testbeds, certification and infrastructure:** National metrology institutes provide test and evaluation services for prototypes and new products based on quantum technologies. Metrology institutes can certify hardware performance, providing companies, governments and investors with a trusted basis for decisions on where to allocate resources. The United Kingdom's [National Physical Laboratory](#), for example, provides capabilities and infrastructure that aren't commonly available to companies, enabling testing that would otherwise require expensive equipment such as dilution refrigerators. Similarly, [Qu-Test](#), a European network of research and metrology institutes, assists industry actors in developing their components and systems for quantum technologies. The initiative provides testing, validation and benchmarking for quantum chips, cryogenic assemblies and photonic sources/detectors, among other devices. Germany's Quantum Computing Initiative provides industrial partners with access to cleanrooms,² workshops, laboratories and offices, all of which are offered free of charge when tied to an industry contract for building a quantum computer or enabling subsystems.

Hybrid approaches that combine quantum processors with classical high-performance computing (HPC) can help bridge the gap until quantum computing reaches sufficient maturity. Several HPC infrastructures are being extended to host quantum computers alongside supercomputing clusters, allowing researchers to test applications in environments that mirror future integrated workflows. An interviewee from a technology company stressed that such hybrid solutions are crucial, as they help reduce hesitation among potential users and make it easier for firms to engage with the technology incrementally. Germany's Quantum Computing Initiative is also providing access to HPC and quantum resources in parallel, enabling companies to co-design and trial workflows without restrictive contractual or intellectual property hurdles. Interviewed technology providers anticipate that breakthroughs in areas such as materials science will depend on "a combination of quantum, AI and HPC," rather than quantum alone.

3.2 Limited awareness of quantum computing and understanding of its implications

Infographic 2. Obstacle overview and related support measures



Low levels of awareness and understanding around quantum computing's actual capabilities and timelines pose another major hurdle. Even in sectors where engagement might be expected, firms often have not

begun preparing because they lack the necessary knowledge base. Firms tend to postpone action, viewing quantum computing as something to address in the future. They may acknowledge the importance of quantum and express interest in its future relevance, yet they often stop short of acting on this recognition to build a deeper understanding of the technology, develop concrete strategies or engage in structured planning. When companies begin to engage with the technology, they often ask very practical but fundamental questions, such as whether it works, whether it can perform better than their current methods, how much it would cost in both the short and long term, and where they would be able to access reliable solutions. Many firms focus solely on the future prospect of fault-tolerant quantum computers, dismissing the potential of currently available and emerging devices in sectors such as chemistry. Interviewees attributed limited awareness to three main explanations:

- At present, most of the examples of potential applications of quantum computing originate from technology providers, particularly start-ups, who promote a wide range of possibilities. However, beyond a handful of promising areas, such as materials science, many firms that may ultimately apply these technologies do not yet have a clear understanding of use cases that detail how quantum computing can be integrated into their specific industry contexts.
- Without access to usable systems, such as mid-scale quantum computers that businesses can experiment with, firms struggle to explore potential use cases meaningfully.
- Awareness tends to be concentrated in just a few individuals within the organisation, leaving most staff and decision-makers without a clear understanding of what quantum adoption would involve.

To raise awareness, many of the organisations interviewed publish and promote resources such as quantum computing readiness assessments, ecosystem studies and other reports that share evidence-based insights with a broad audience. For example, [QURECA](#) has prepared readiness assessments and ecosystem research, and has developed Quantum Technology Readiness Levels to benchmark preparedness among different stakeholders. The [Quantum Technology and Application Consortium \(QUTAC\)](#), a German industry consortium of 14 of the country's leading corporations, maintains the [Quantum Computing Monitor](#), a national ecosystem study tracking national progress in the application of quantum computing. National bodies, such as the UK's National Physical Laboratory, publish white papers, studies and roadmaps to offer guidance for firms, particularly SMEs that often lack the resources to evaluate quantum opportunities independently.

Alongside publicly available reports and consultancy outputs, several organisations host networking opportunities to help firms engage with quantum computing. These initiatives expose non-specialist decision-makers and technical teams to key concepts and emerging industry practices, helping them understand where quantum computing may become relevant to their organisations. For instance, the United States' [Quantum Economic Development Consortium \(QED-C\)](#) has organised sector-focused workshops on applications ranging from electric grid management to biomedical imaging, making the outcomes publicly accessible as reports. In Japan, the [Quantum Strategic Industry Alliance for Revolution \(Q-STAR\)](#) runs workshops for both engineers and executives, ensuring that firms build awareness not only at a technical level but also at a managerial level. Innovate UK's webinars and workshops, as well as the Australian government's "[Quantum Meets](#)" (organised by the country's Chief Scientist in partnership with the [CSIRO](#) and other organisations), link technology providers with prospective end users to build awareness of potential applications.

Technology providers and consulting companies offer business advisory services that help firms engage with quantum computing. These services help companies evaluate their current capabilities, identify gaps and plan activities for building quantum readiness. They include, for example, producing executive briefings, organising tailored workshops and conducting readiness assessments. Advisory support often provides sector-specific examples and application roadmaps, enabling executives to connect emerging technologies to their own business context. This support helps firms contextualise quantum developments within their own industries, understand risk exposure and potential use cases, thereby taking first steps

towards building readiness. In addition to raising awareness among firms, promoting awareness and acceptance within civil society is equally necessary to build trust in quantum technologies (Box 5).

Building readiness cannot be fully outsourced. While technology providers and other actors may provide valuable advisory, firms must develop internal capacity to understand how quantum computing could impact their industry, operations and products. This involves dedicated staff and resources to explore applications, including by experimenting with algorithms, conducting feasibility studies and engaging actors in quantum technology ecosystems. Beyond identifying potential use cases, companies also need to assess how quantum computing could integrate with existing digital and operational systems. As one technology provider noted, commercial applications depend on a complete technology stack, including hardware, software, middleware and control systems. Firms need in-house capacity to evaluate how these layers could be adapted for day-to-day operations.

Box 5. Raising awareness and acceptance of quantum technologies in civil society

Engaging citizens plays an essential role in building trust and fostering informed debate around quantum technologies. Public awareness initiatives help demystify complex concepts, enabling people to better understand not only the potential benefits but also the limitations, risks and ethical implications. Excessive hype can undermine trust, as unmet expectations may lead to scepticism and disengagement.

Transparent and open communication about both the limitations and opportunities, as well as the realistic timelines of quantum development is therefore needed. Public engagement in the development and use of quantum technologies fosters inclusiveness and accountability, ensuring that these technologies align with broader societal needs and values. Social acceptance also matters for business uptake: firms are more likely to invest in and deploy quantum solutions when there is a supportive societal context that reduces reputational risks, attracts talent and creates demand for quantum-enabled commercial applications.

Source: OECD (2025^[1]).

Public narratives, especially in the media, often portray quantum computing as a near-term revolution that will transform entire industries within just a few years. In practice, however, progress is slower and the benefits are likely to emerge first in specific, narrowly defined application areas rather than broadly across industries. Interviews suggest a gap between academic caution and the more optimistic narratives directed at prospective end users: While researchers tend to emphasise the limits of current capabilities and the long timelines still ahead, many companies are instead exposed to industry reports that present a far more confident outlook. These reports often highlight ambitious applications and forecast shorter development horizons, which can create unrealistic expectations within firms.

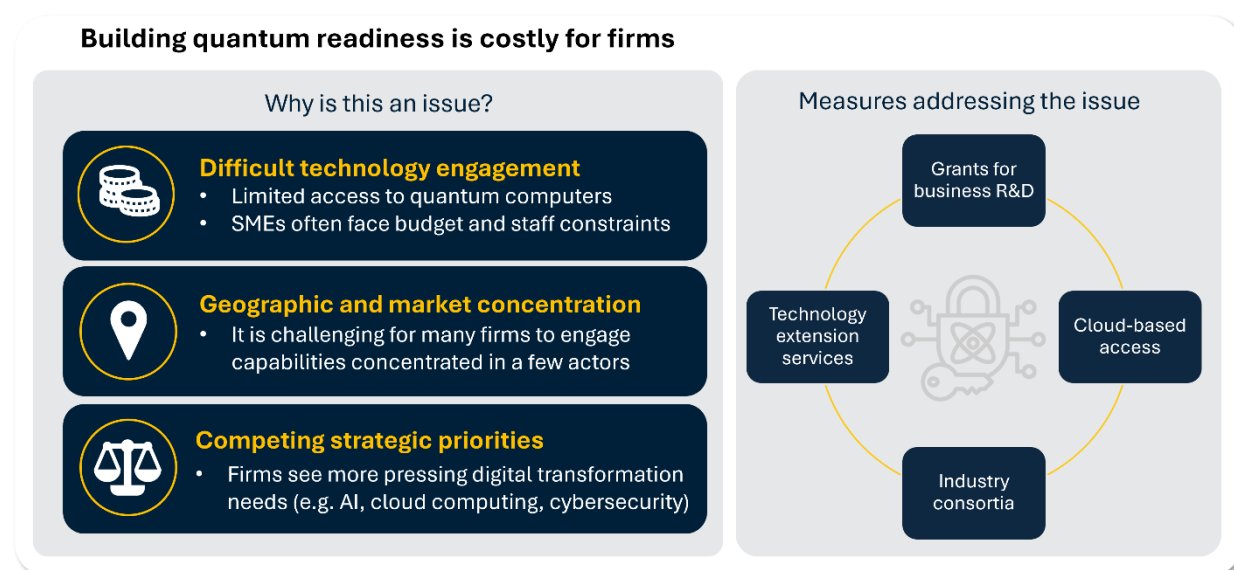
Excessive hype regarding quantum computing capabilities can lead to unrealistic near-term expectations, leaving businesses vulnerable to strategic missteps and, ultimately, disillusionment and mistrust when progress proves slower. Companies may conflate quantum advantage with immediate business utility or lack clarity on how quantum computing differs from other emerging technologies, such as AI or high-performance computing. Internal awareness is often limited to a small group of technical experts, while strategic leadership lacks the necessary knowledge to make informed decisions on how to invest.

To address this challenge, organisations supporting readiness take care to explain that quantum computing remains at an early stage of maturity and that firms should not expect near-term business improvements or production-ready applications. Instead, they frame readiness investments as exploratory, supporting long-term learning. The most valuable outcomes at this stage include (i) training staff, (ii)

building familiarity with quantum algorithms and software, and (iii) running small proofs of concept to test basic methodologies and explore potential business applications. As the field is evolving rapidly, readiness is less about implementing stable systems and more about gaining early exposure and developing internal capability to adapt as the technology matures.

3.3 The cost of building readiness

Infographic 3. Obstacle overview and related support measures



Until recently, access to a quantum computer was considered prohibitively expensive for most prospective end users, since it required direct acquisition. In addition to the significant upfront investment, companies need to invest in specialised personnel to operate and maintain the machines, along with extensive infrastructure such as dedicated facilities with controlled environments and technical safeguards. Given these substantial and recurring costs, very few firms can realistically pursue this model.

Cloud-based platforms are increasingly facilitating access to quantum compute time on remote machines. This model makes experimentation with quantum computing more accessible, shifting both the financial and technical burden away from the end user and enabling firms to participate in building readiness without committing to costly infrastructure. While this approach thereby significantly lowers the entry barrier, the cost of cloud-based access is not negligible. According to one estimate, using a remote quantum computer for 12 hours can cost approximately USD 70 000 (OECD, 2025^[1]).

Even under a cloud-based access model, businesses still face substantial costs when preparing for quantum computing. These include hiring or training personnel, conducting internal readiness assessments, building awareness across departments and experimenting with quantum-inspired algorithms or pilot use cases. For many companies, especially those outside the technology sector, these activities compete with more immediate digital transformation priorities, such as AI integration, cloud migration or cybersecurity upgrades. In addition, access to quantum computing through cloud services is subject to the terms and conditions set by providers, which may be affected by export control regulations. As a result, availability of quantum cloud services can be limited for users residing in, or connecting from, certain jurisdictions. These regulations particularly affect firms in emerging economies (AWO, 2024^[35]).

Given the uncertainty around when and where quantum computing could deliver a competitive advantage, many firms are reluctant to commit resources to a technology with unclear short-term returns. This limits engagement, even for firms with the means to invest. As Diego Quiñones, Innovation Lead at Innovate

UK, observes, many organisations adopt a “wait-and-see” approach, slowing readiness efforts despite acknowledging the technology’s long-term potential. For SMEs, limited financial capacity and staff resources make experimentation with quantum technologies especially difficult.

A related challenge is the economic and technological concentration observed in a small number of developed countries (OECD, 2025^[1]). The high upfront investment and specialised expertise required to develop quantum computing have created a landscape dominated by a few key countries and firms. The concentration makes it difficult for many companies to access the knowledge and partnerships needed to build quantum readiness. Firms in emerging economies and countries with modest investments in quantum technologies face particular hurdles due to limited access to technology.

To ease the financial burden of early experimentation and encourage investment and cross-sector partnerships, governments have introduced dedicated programmes that provide grants for business R&D. The programmes described below support projects spanning all quantum technologies, including quantum computing:

- In the United Kingdom, the [Commercialising Quantum Technologies Challenge](#) (Innovate UK) funds, among other activities, collaborative projects that bring together industry and research partners to develop quantum products and services. Beneficiary firms are required to co-invest, with participants also gaining access to academic expertise and testbeds. The scheme supports Collaborative Research and Development demonstrators, which focus on building quantum devices (including computers) that progress to system-level trials with end users. The programme also funds feasibility studies and early-stage projects that advance user-defined quantum products, services, devices, components or supply chain elements. Computing-focused projects span application areas such as drug discovery, advanced materials (including batteries), industrial and transport optimisation and quantum machine learning for finance and genome sequencing.
- In Australia, the [Critical Technologies Challenge Program](#) (Department of Industry, Science and Resources) operated in two stages. In stage one, feasibility grants enable applicants to propose solutions to nationally significant challenges and carry out detailed projects to test and demonstrate technical viability. In stage two, applicants are invited to apply for demonstrator grants, which support the transition from feasibility results to working prototypes or demonstration systems. The programme was broad in technology scope but included targeted quantum computing projects, for example on optimising the performance of energy networks, as well as improving transport routes, logistics and supply chain operations.
- In Germany, the Federal Ministry of Research, Technology and Space (BMFTR) has mandated VDI Technologiezentrum to support the Ministry in designing and administering national funding programmes for quantum technologies. A key feature of programmes is their emphasis on early industrial involvement, ensuring that collaborative consortia include companies alongside universities, start-ups and research organisations. Project formats range from large-scale R&D collaborations and technology testbeds to smaller schemes such as researcher exchange programmes and education-focused initiatives (including the distribution of low-cost one-qubit “makerspace” quantum computers for schools). An overarching goal is to lower barriers for firms to engage in joint R&D, stimulate private investment and accelerate the transition of quantum technologies from academic research to industrial application.

In addition to grants for business R&D, technology providers and government-backed initiatives are facilitating cloud-based access to quantum computing resources:

- Technology providers increasingly offer access through cloud platforms. These services enable firms to access quantum hardware remotely, often in conjunction with development toolkits and expert support.

- Germany's Quantum Computing Initiative seeks to provide quantum computing resources to project partners while aiming to ensure that participating companies retain their intellectual property. This approach contrasts with typical commercial cloud arrangements, which restrict companies' ownership of information shared with service providers, including novel algorithms. For firms engaging in early-stage experimentation, such unfavourable intellectual property conditions are perceived as a significant barrier.

Several organisations support companies in exploring business-specific use cases through collaborative platforms and technology extension services, lowering entry barriers and creating opportunities for firms to conduct collaborative R&D, feasibility studies and business-specific experimentation that would otherwise be too costly or inaccessible. For example:

- QUTAC seeks to provide its members with shared access to quantum computing platforms and brings together their internal quantum experts. The cross-company QUTAC teams are researching quantum algorithms and industrial use cases. Because the technology is still in its early, pre-competitive stages, participating firms are willing to share knowledge and R&D results, helping one another overcome common challenges. This collaborative model aims to spread the cost and risk of experimentation and has already led to the identification and documentation of viable quantum computing use cases across different industries (QUTAC, 2021^[36]).
- At the Dutch research institute [QuSoft](#), two different models have been developed to support firms in exploring quantum computing. On the one hand, companies may engage in longer-term collaborations by co-funding a PhD or postdoctoral researcher, thereby embedding quantum expertise into projects that span several years and allow for a deeper exploration of complex problems. On the other, the [Quantum Application Lab](#) offers a shorter, more applied format that typically spans only a few months up to a year. These projects are designed to sit between academic research and consultancy, providing firms with tangible outputs, such as proofs-of-concept (including prototype code) and recommendations for building a quantum computing roadmap.
- Technology providers themselves can also complement cloud-based access with R&D collaboration, by assigning staff time to co-develop algorithms and use cases with their clients. This support helps accelerate capability building by embedding provider expertise directly within the client company.

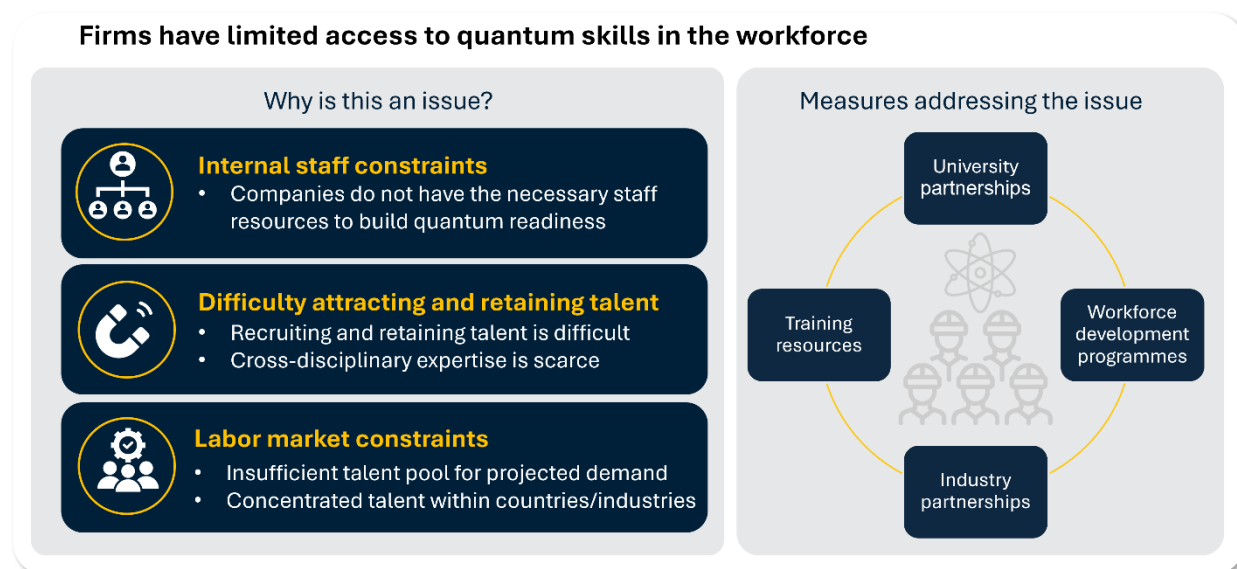
For most firms, it may be premature to commit significant resources to quantum computing, but staying informed of developments remains valuable. As Tim Prior, Quantum Programme Manager at the UK's National Physical Laboratory, explains, being “quantum curious” means actively monitoring both the opportunities and risks that may emerge, even while the technology is not yet ready for adoption. Cultivating this awareness enables companies to anticipate when applications become relevant, reducing their risk of being caught off guard by faster-moving competitors.

3.4 Limited access to skills in the workforce

Readiness requires staff who combine computer science and quantum knowledge with industry-specific expertise, yet such cross-disciplinary skills are rare. Without individuals who can bridge these areas, firms will continue to struggle to progress beyond general awareness into more advanced stages of readiness. As Celia Merzbacher, Executive Director of QED-C, notes, most companies lack internal advisors with the capacity to assess the potential relevance of quantum computing. Hiring even a small number of experienced staff (two to four employees, according to one interviewee) can provide a foundation for organisational learning, enabling firms to experiment with algorithms, run proof-of-concept projects and engage with the quantum ecosystem. However, companies face persistent challenges in finding and retaining the right expertise. Job board data and other sources already point to a growing shortage of

quantum talent (OECD, 2025^[1]). Beyond quantum-specific specialists, companies also need support from data scientists, technicians and other complementary profiles.

Infographic 4. Obstacle overview and related support measures



Another difficulty concerns how knowledge is distributed within companies. Some firms employ highly trained quantum specialists, including PhDs, but this expertise is often concentrated in just one or two individuals. The rest of the organisation may have little to no awareness of the subject, making preparedness fragile and overly reliant on those experts. Should they depart, the organisation's ability to engage with quantum computing could quickly erode.

According to a technology provider, there is often a misconception about the type of skills needed. Many companies assume they need to recruit large numbers of quantum physicists. In reality, the quantum workforce pyramid is inverted: Only a small number of physicists are required at the top, while the majority of roles involve engineers, system specialists and other applied experts who can build and integrate quantum systems with existing technologies, including computer scientists and IT specialists who can develop the necessary software stacks. Interviewees observed a mismatch between the skills industry needs and those nurtured by universities. Caroline Bürsgens, Manager of QUTAC, observes that Europe has built up considerable strength in basic quantum research through its universities and scientific institutes, especially in the field of quantum hardware. However, the real challenge for firms lies in transitioning from this research base to applied work that yields practical results. In particular, this also requires expertise on the software side. The most pressing weakness lies in the area of quantum algorithms and their use cases, where a pronounced shortage of expertise exists. Another challenge is finding individuals who can bridge quantum and classical computing systems. Firms increasingly need engineers and developers who can work at the interface between emerging and conventional information technologies. These gaps are described as significant enough to threaten competitiveness, as developing the right algorithms will ultimately determine whether companies can effectively utilise quantum computing for their industries.

The uneven distribution of skilled quantum workers across countries and sectors also constrains access to skills. While global players in selected countries have assembled large teams of algorithm specialists, in other countries, industry players tend to rely on smaller groups, resulting in projects that progress more slowly. Some sectors, such as finance, are relatively well-positioned because they already employ staff who are deeply familiar with algorithm development and related methods. While these employees may not

yet be specialists in quantum computing, they have a foundation that could be built upon through vocational learning and training. By contrast, firms in sectors like chemical manufacturing are often at a disadvantage. They typically lack comparable internal expertise, leaving them with limited ability to engage meaningfully with quantum computing. Without in-house skills or partnerships with universities and other actors in the technology ecosystem, these companies risk falling behind as quantum computing matures. This creates a two-speed environment: countries and industries with strong technical workforces can begin exploring quantum readiness, while others face significant barriers simply because they lack the people who can bridge the gap between current knowledge and quantum-specific requirements. Unless more emphasis is placed on developing and attracting skilled experts, firms in lagging regions and sectors may find themselves at a disadvantage when the technology matures.

While interviewees noted that the quantum talent pipeline is expanding, much of the growth is in academic profiles whose training does not always align with practical industry needs. Many experienced researchers move to attractive career prospects in industry, tightening workforce capacity in university groups and making it harder for firms without substantial internal resources to access expertise through academic collaborations. Interviewees observed a clear lack of educational pathways and certification programmes for industry-relevant skills, which adds to the overall uncertainty faced by companies. Only a handful of graduate programmes focus on quantum engineering or software development, and these are concentrated in specific regions, further reinforcing geographical imbalances. Addressing these challenges is a long-term task that involves not only increasing the supply of researchers but also tailoring education and training pathways to meet companies' diverse needs.

Given the difficulties in hiring and nurturing in-house talent, expertise often comes from technology providers. Whether through on-premise installations or cloud access, providers play a central role in running the systems and supporting operations. This reliance is especially pronounced in on-premise deployments: Although a machine may be installed at a customer site, provider teams typically remain on-site to operate and maintain the system. Client companies allocate only limited staff to oversee technical aspects. The complexity of such installations reinforces this dependency, as they require specialised infrastructure and equipment, including dedicated facilities such as cleanrooms, refrigeration systems and strict security and technical conditions. Even under cloud-based access models, technology providers frequently develop tailored quantum algorithms and co-build use cases with firms.

Partnerships with universities and research centres are seen as a valuable channel for identifying and recruiting talent. QuSoft, for example, contributes directly to education by ensuring that university students have a strong background in fundamental tools needed to understand and develop new algorithms, with course contents aligned with industry needs. The institute aims to ensure that students not only understand the algorithms known today, but that they can also actively develop and work with algorithms that will emerge in the future. Engagement with such public research organisations provides early visibility of the talent pipeline as well as opportunities for collaborative R&D. However, formal research collaborations are financially demanding, since they usually involve hiring a PhD student or postdoctoral researcher, with costs typically exceeding EUR 100 000 per year in the Netherlands, for example, according to Dr. Koen Groenland, Quantum Innovation Officer at the University of Amsterdam and QuSoft. For many firms, especially those without a strong R&D culture, this is a significant barrier. Banks, for example, were described as generally lacking the tradition of funding such academic-style work, while companies with established research arms are better positioned to afford it.

Consultancies often collaborate with universities by bringing in academic experts to co-develop and deliver advanced training content, enabling companies to access cutting-edge knowledge while also helping to identify future potential recruits. At a policy level, interviewees also highlighted programmes supporting industry-university exchanges and collaborative research projects, where researchers spend time embedded in companies, as a form of support that strengthens both innovation and the talent pipeline.

Several actors provide tailored upskilling programmes that help companies progress from raising awareness to nurturing expertise in staff:

- QURECA delivers skills development at different education levels, offering online, in-person and live-virtual formats and assembling modular packages for managers and technical staff (e.g. “quantum computing for finance and cybersecurity”), adjusted in duration and depth to client needs. For advanced topics, QURECA partners with domain experts to deliver content. As most companies are currently at an awareness-building stage, QURECA’s role is often to help identify which employees to upskill and to design a combined training and strategy plan.
- [Quantum.Amsterdam](#), an innovation hub founded by QuSoft, CWI Amsterdam and the University of Amsterdam, provides professional training sessions that companies can book on-site, typically as focused half-day workshops that introduce concepts or provide practical, team-based learning. These sessions are complemented by open public events that explain the basics of quantum computing to a broader audience, as well as visits to start-ups that showcase concrete applications.
- Q-STAR provides structured training opportunities for both engineers and executives, combining introductory courses that build basic literacy with more advanced sessions for specialists. The consortium not only runs its own programmes but also encourages member companies to organise internal training, fostering capability-building that is embedded within firms themselves.
- Technology providers also embed skills development into their commercial readiness programmes and by forming partnerships with clients in which access to hardware or cloud services is combined with training and co-development of use cases. Providers use a mix of hands-on workshops, scenario-based exercises and strategic guidance, designed as iterative learning journeys rather than one-off events. This support situates training within the context of a firm’s own computational challenges, allowing engineers and managers to develop practical expertise while exploring concrete applications. In some cases, developers place experts within companies regularly to co-develop algorithms and provide operational support. In others, firms are trained to access and use remote platforms under the guidance of the provider.

To complement formal training programmes, organisations are developing a range of publicly available resources to support workforce upskilling. Several interviewees highlighted the value of training documentation provided by technology providers for their software and hardware, which enables end users to self-learn and integrate tools into existing workflows. Alongside these materials, hackathons embedded in regional conferences combine outreach with practical training. They engage students and early-career professionals while giving firms visibility into emerging talent. In addition, the availability of open-source tools was noted as a helpful resource for experimentation, offering companies low-cost opportunities to familiarise staff with quantum programming and algorithm development.

The shortage of skills in quantum technologies is not only a current obstacle but also a growing risk for the future. Interviewed organisations stress that the workforce is too small to meet the demand that would arise once the technology matures. If efforts to train and upskill people do not begin now, the gap between the number of professionals needed and those available could widen dramatically over time. One interviewee drew a parallel with artificial intelligence, where the demand for expertise has quickly outstripped the supply. Without early action, quantum computing could face an even more acute shortage, leaving firms unable to take advantage of the technology when it becomes viable for broad commercialisation.

4. Outlook

Companies preparing for quantum computing understand the technology’s current limitations and uncertainties, yet treat readiness as a strategic investment. Early preparation enables them to closely track

developments and respond quickly as the technology matures, creating the potential for a first-mover advantage over less prepared competitors. In practice, most firms pursue readiness incrementally, starting with low-cost activities that build awareness and internal capacity. These may include awareness events to highlight emerging technology capabilities and opportunities, training sessions led by external experts or exploratory workshops to gauge potential business-specific applications. Such activities serve as stepping stones, helping firms assess the relevance of quantum technologies to their operations and determine whether deeper engagement is justified. More advanced readiness requires targeted investment in business-specific use-case exploration (e.g. feasibility studies and proofs-of-concept). This investment enables firms to test algorithms, examine integration with existing systems and generate evidence of where quantum computing could deliver a genuine competitive advantage.

The evidence gathered in this paper suggests that readiness efforts today are unevenly distributed. Large firms with substantial R&D budgets are the primary early movers, motivated by strategic considerations and a concern about being left behind on a transformative technology. By contrast, many other firms, including those acknowledging the potential of quantum computing, delay engagement due to cost, skills shortages, unclear business cases and technology immaturity. Technology capabilities tend to be concentrated among large firms in technologically leading countries and in certain industries, where some sectors are better placed to build readiness while others may lag behind. These gaps raise the risk of a widening divide between early adopters and the broader digital economy. Unless targeted support mechanisms address these barriers, quantum readiness may remain concentrated in a narrow group of firms, thereby limiting its broader economic benefits.

Organisations supporting readiness are already helping to bridge these gaps, yet interviewees stressed that current efforts need recalibration and expansion. Awareness-raising remains essential, particularly in addressing what one participant called the “final mile problem,” where high-quality reports and resources often fail to reach the firms that most need them. Financial support instruments may also need to be adapted, ensuring that grant schemes lower barriers for a broader range of firms, including SMEs, and that they are designed with stronger input from industry to maximise uptake and practical impact.

Looking forward, readiness efforts will need to evolve alongside the technology itself. Current priorities, such as building awareness, providing access to cloud-based hardware and running proofs-of-concept, must be complemented by forward-looking support for software development, application design and workforce training. Business associations across different countries underscored the urgency of expanding workforce training pipelines to ensure firms can apply quantum computing once the technology matures. Likewise, technology providers highlighted the importance of interoperability between quantum, AI and high-performance computing, recognising that hybrid approaches will likely be the most practical entry point for industry in the near term. Regulation and compliance will play an increasing role as commercial deployment expands and business use cases mature.

Readiness efforts will continue to be shaped by broader strategic and geopolitical considerations, with some organisations signalling that forthcoming revisions to national strategies may provide the opportunity to refresh priorities and chart new directions. Ensuring that such shifts translate into accessible, practical support for a wide range of firms will be critical to prevent readiness from remaining limited to a few early movers and to enable broader participation in the emerging quantum computing economy.

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Notes

¹ See <https://www.imf.org/en/publications/weo/weo-database/2023/april/groups-and-aggregates>.

² A cleanroom is a specially controlled environment designed to maintain very low levels of dust, airborne particles, microbes, chemical vapours, and other contaminants.

Annex A. Interview questions

Part 1. About your organisation

1. How does your organisation support companies building readiness for quantum computing?
2. Roughly how many firms do you work with, if it is possible to say?
3. Do you have a specific strategy to find and reach out to the population of enterprises most likely to adopt quantum technologies? How does that work?

Part 2. Barriers to quantum computing readiness

1. The impact and contributions of quantum computing have not yet been completely discovered. To what degree do you think that firms, in general, are aware of the potential uses of quantum in their sector? And what do you consider key barriers to starting quantum readiness efforts today?
2. Do you think that firms underestimate the difficulty of trying to build capacity for quantum computing? If so, how?

Part 3. Effective forms of support

1. One problem we find is that it is currently not possible to calculate the return on investments in quantum computing. What motivates firms to invest anyway?
2. In your experience, what types of advice and support are most effective for enterprises trying to prepare for quantum computing?
3. In the future, is there something different with respect to supporting quantum preparedness that you plan or would like to implement?